

Module 3 – Machine Learning

1. Machine Learning Introduction

What is Machine Learning?

Machine Learning is a method where computers learn from data and make predictions or decisions without being explicitly programmed.

Why It Matters

- Automates decision-making
 - Finds patterns in data
 - Powers modern AI systems
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2. Supervised Learning

What is Supervised Learning?

A type of machine learning where models are trained using labeled data.

Examples

- Predicting house prices
- Email spam detection

Key Idea

Input data + correct output → model learns mapping

3. Unsupervised Learning

What is Unsupervised Learning?

A type of learning where the model works with unlabeled data.

Examples

- Customer segmentation
- Pattern detection

Key Idea

Model finds hidden structure in data

4. Model Training

What is Model Training?

Training is the process of feeding data into a model so it can learn patterns.

Steps

- Provide dataset
- Train model
- Adjust parameters

5. Evaluation Metrics

What are Evaluation Metrics?

Metrics are used to measure how well a model performs.

Common Metrics

- Accuracy
- Precision
- Recall
- F1 Score

6. Key Concepts to Remember

- Machine learning learns from data
- Supervised learning uses labeled data
- Unsupervised learning uses unlabeled data
- Training helps model learn patterns
- Metrics evaluate performance

7. Summary

Machine learning is the core of data science. By understanding supervised and unsupervised learning, model training, and evaluation metrics, you can build intelligent systems that make accurate predictions.